Project Title : Adventura: Plan Your Trips

Table of Contents

Section No	Content	Page No
1	Introduction	3
2	Functional Requirements	4-5
3	User Manual	6-17
4	Frontend Development	18-20
5	Backend Development	21-24
6	Technology (Framework, Languages)	25-27
7	Github Repo Link	27
8	Individual Contribution	27

Introduction

Adventura: Plan Your Trips is a well functional and easily accessible website for the travel enthusiasts who are willing to plan and organize their holidays by arranging the perfect travel plan for them.

Nestled in the heart of South Asia, Bangladesh is a land of diverse landscapes, rich history, and vibrant culture that beckons travelers to embark on an unforgettable journey. Often overshadowed by its more prominent neighbors, this enchanting country is a hidden gem waiting to be discovered by intrepid explorers seeking unique experiences.

To make the travel in this beautiful country more efficient, attractive and cost efficient, we introduce you to Adventura. This modern platform will ensure you have the best experience for your travel plans. It will work as a guide and give you some perfect combinations of packages, trip plans, hotel suggestions, place and transport reviews and so more.

After registering you will dive into the website where you can easily navigate to some of the easy accessible features. For example, you can find your favorite spots to visit and get to know in detail by simply clicking one link. You can also find the suitable hotels for the places. Don't forget to put your thoughts in the comment box ! You can also navigate to the trip plan module where you can make your own tour package by submitting some of the requirements. You can also view the cost and make payment for your own package. If you don't want to waste your precious time creating your own package, there is another feature where you can find multiple packages which are ready for you to claim.

This platform will serve the best experience to you for creating a perfect schedule, costing and planning for your travel. This website is the perfect stop for all the travel enthusiasts of Bangladesh.

Functional Requirements

1. Authentication:

- Allow users to create accounts and log in securely.
- Enable users to create and manage their profiles with personal details.

2. Explore Spots:

- Offer a search functionality to explore and discover popular destinations across Bangladesh.
- Provide detailed information about each destination, including pictures, description, transportation by a pop up information box.
- Provide the user's own personal review of particular spots.

3. Trip Planning:

- Users can select destinations (District and Spots) where he/she needs to travel.
- Users can select the transportation way he/she needs to travel to their destination and also find their return transportation at the same page.
- Users can view the hotels available in their desired destination.
- Users can choose a time period of staying in those spots.
- Users can view a detailed trip plan by a pop up.
- Users can view their own plan and proceed to payment or they can delete their plan.

4. Hotel Search:

- Users can search for any kind of hotels for any spots/districts.
- Users can provide their own personal review for the specific hotels.
- Users can view the hotel location by an integrated map.

5. Packages:

- Users can search for packages for any particular spot.
- The detailed package will be listed for user interface.
- Users can Book a package and then can proceed to payment. There will be a waiting function for the user until the Admin approval.

6. Admin Panel:

- Admin can login separately from the user by a specific email and password.
- Admin can update user information.
- Admin can add packages, update packages and delete packages.
- Admin can approve a user’s payment and their package bookings.

User Manual

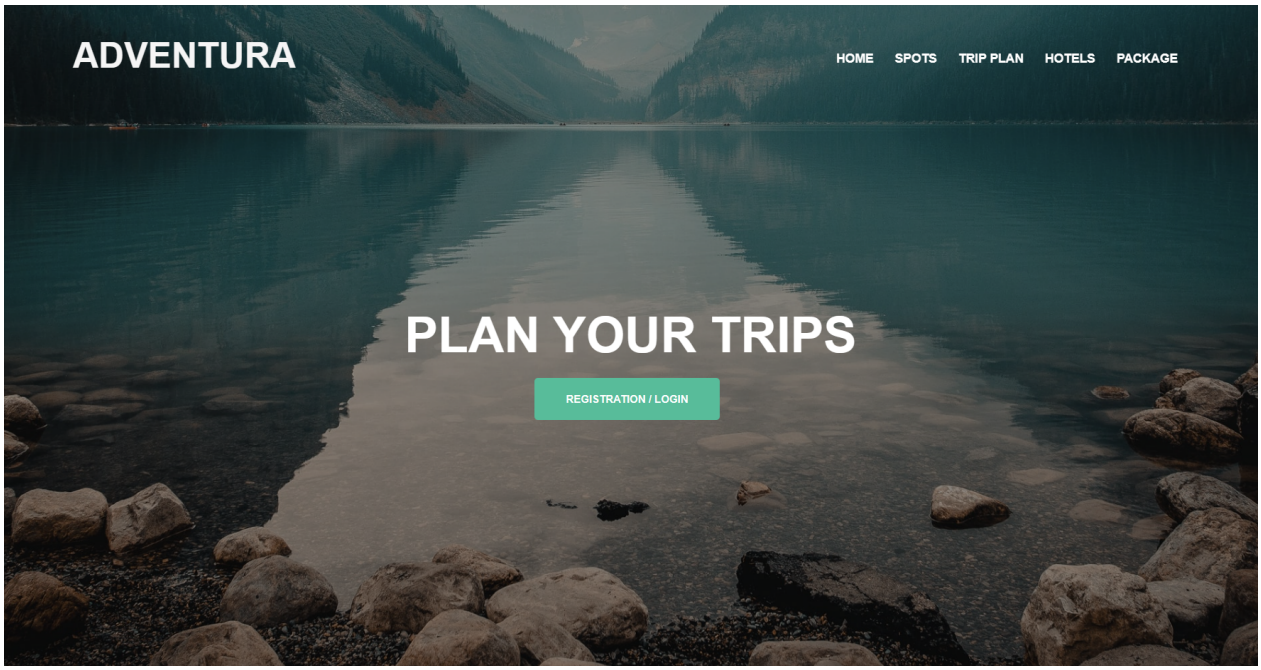
We got two major interfaces in this project. One is user interface and another one is admin interface. So, the description is separated into 2 parts.

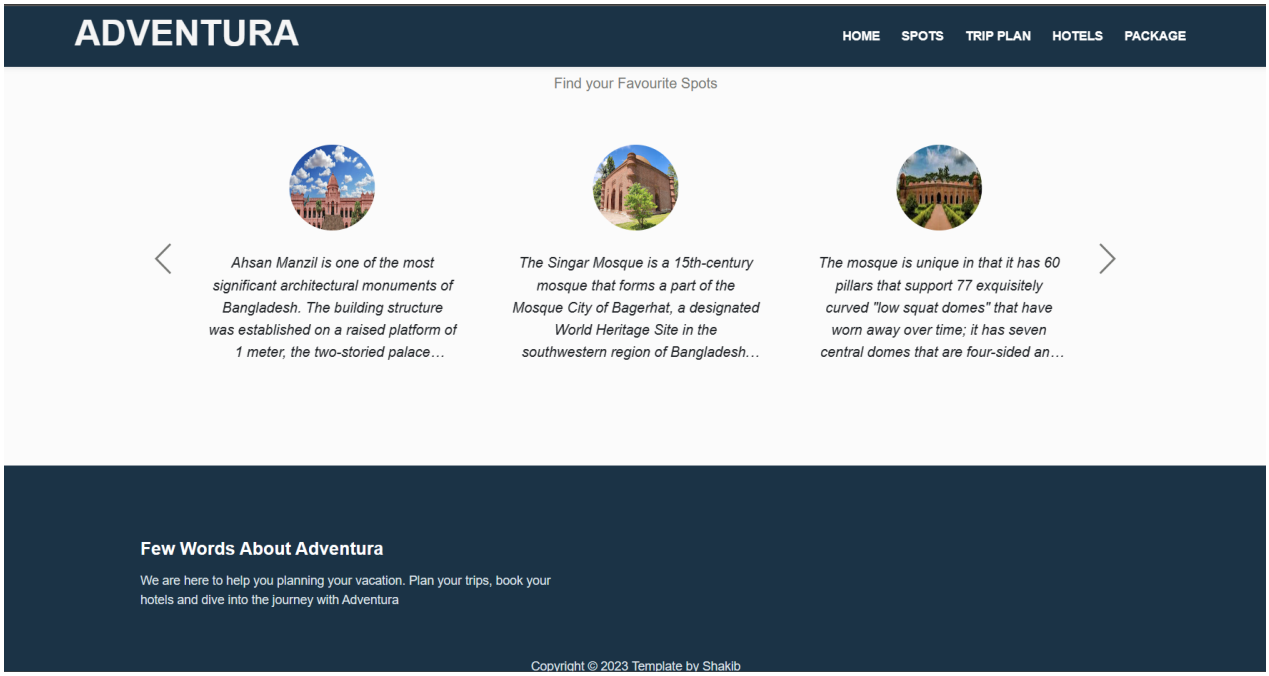
User Interface

There are six major modules for the user side. They are- homepage, search, music player, authentication, registration and personal account of a member.

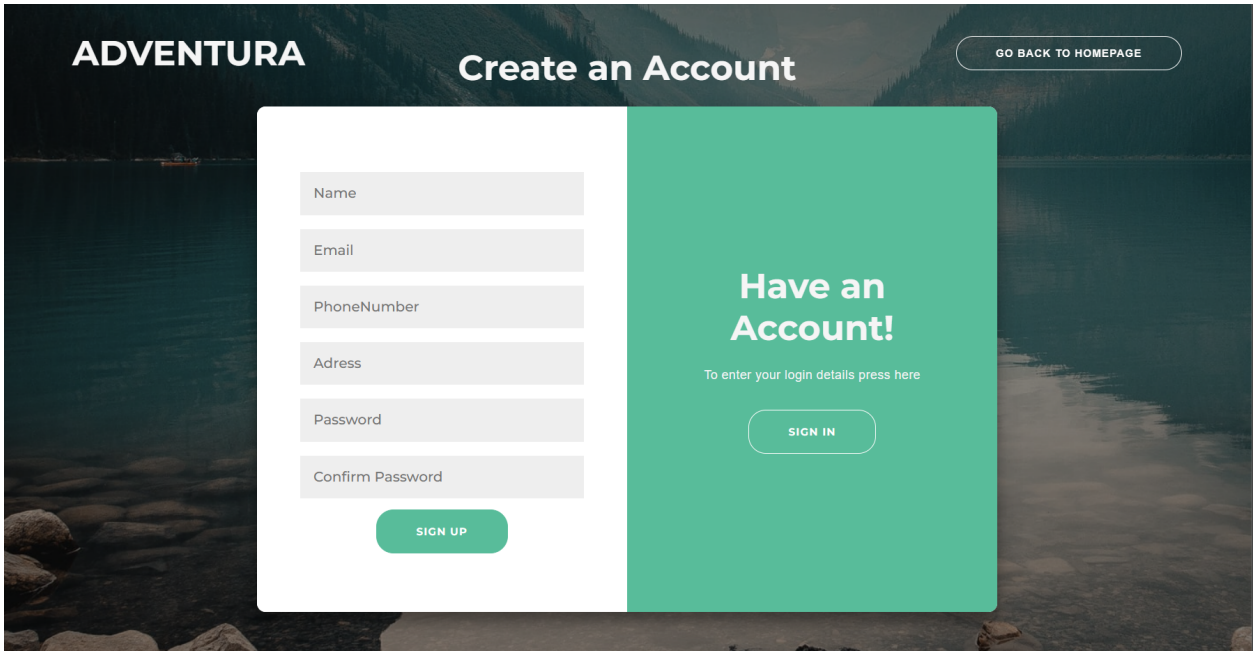
HomePage

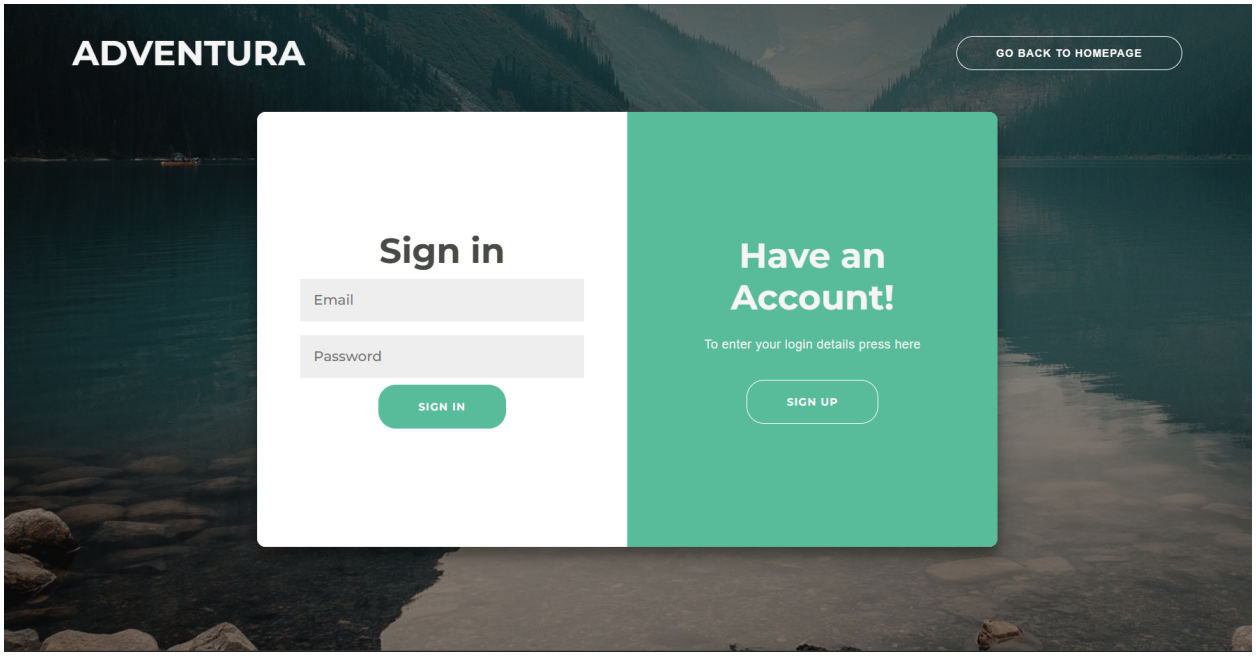
When a user initially accesses this website, both members and non-members receive this homepage. Figure shows the homepage's user interface.
This page has a Login/Registration button and in the bottom of the page users will get reviews of different tourist spots randomly swapping formats.





Registration/Login Button: For using this site, users need to authenticate first. If a user already has a verified account, he/she might login using email and password. And if a new user is there, he/she would need to provide Name, email address, phone number, address and password. The password confirmation part is also needed to be fulfilled.





DashBoard

After successful login/registration, the user will get into the dashboard, where a welcoming text will be shown. There the user will find several buttons-

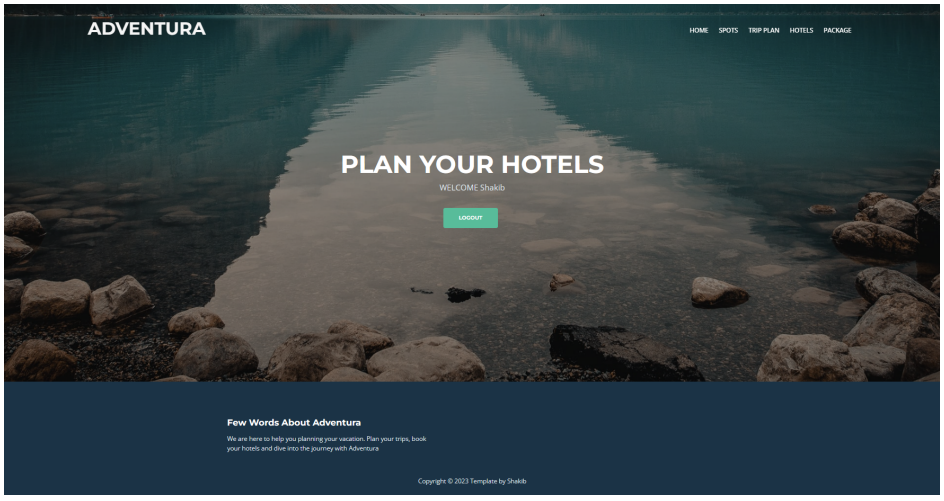
Home: This button is to come back to the dashboard from any other page.

Spots: By pressing this button, the user will get the page to select tourist spots with some suggestions.

Trip plan: This button is to take the user to the planning module.

Hotels: To select hotels or to view the resort options, this button will get into another page.

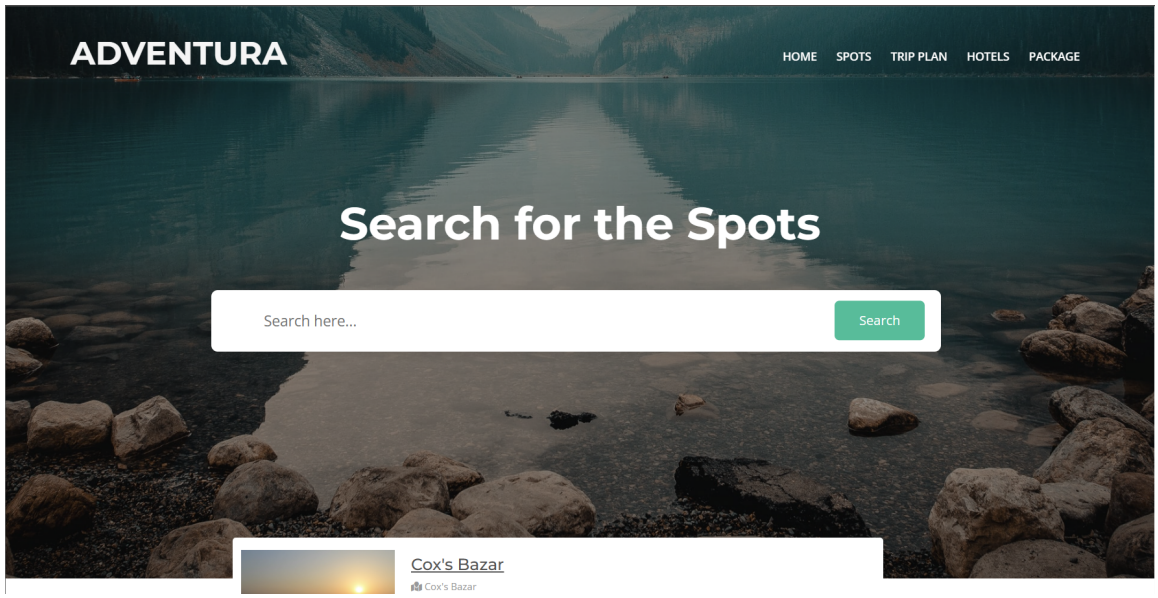
Logout: This button is for logging out from the website.

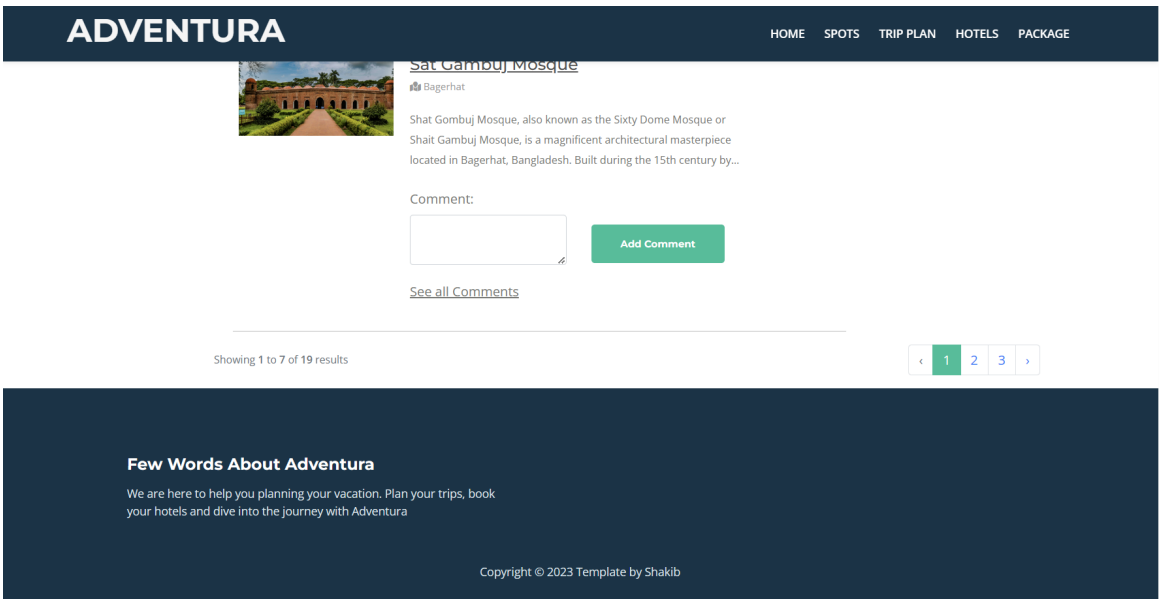


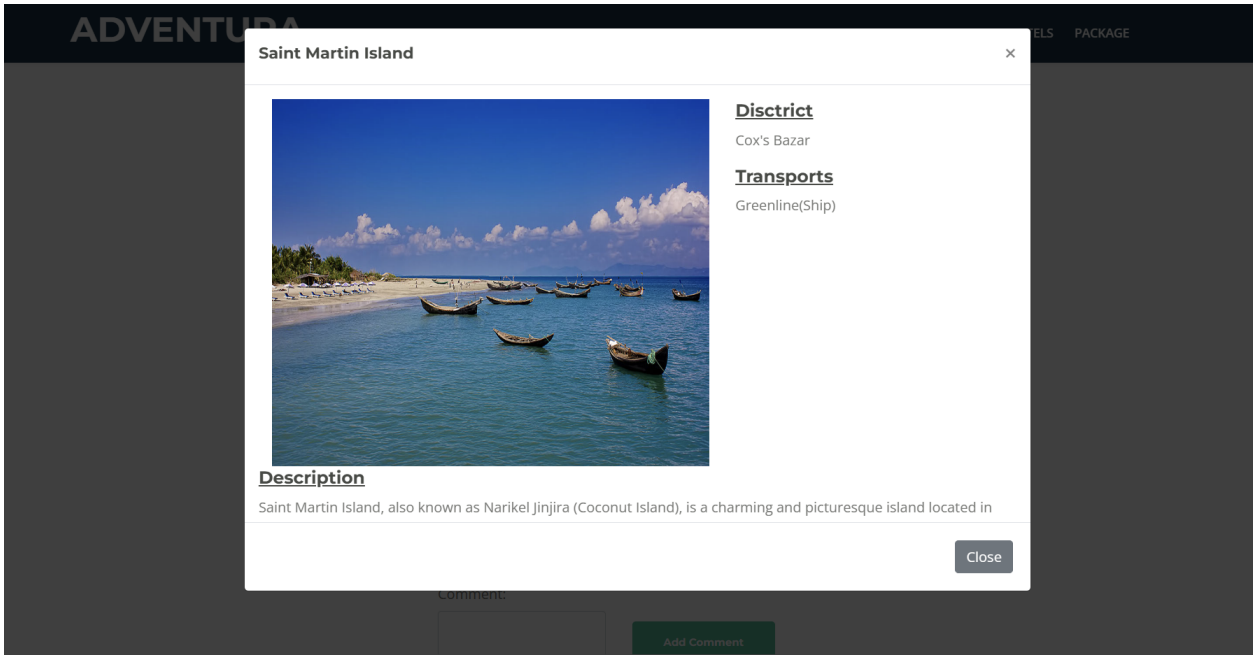
Spots

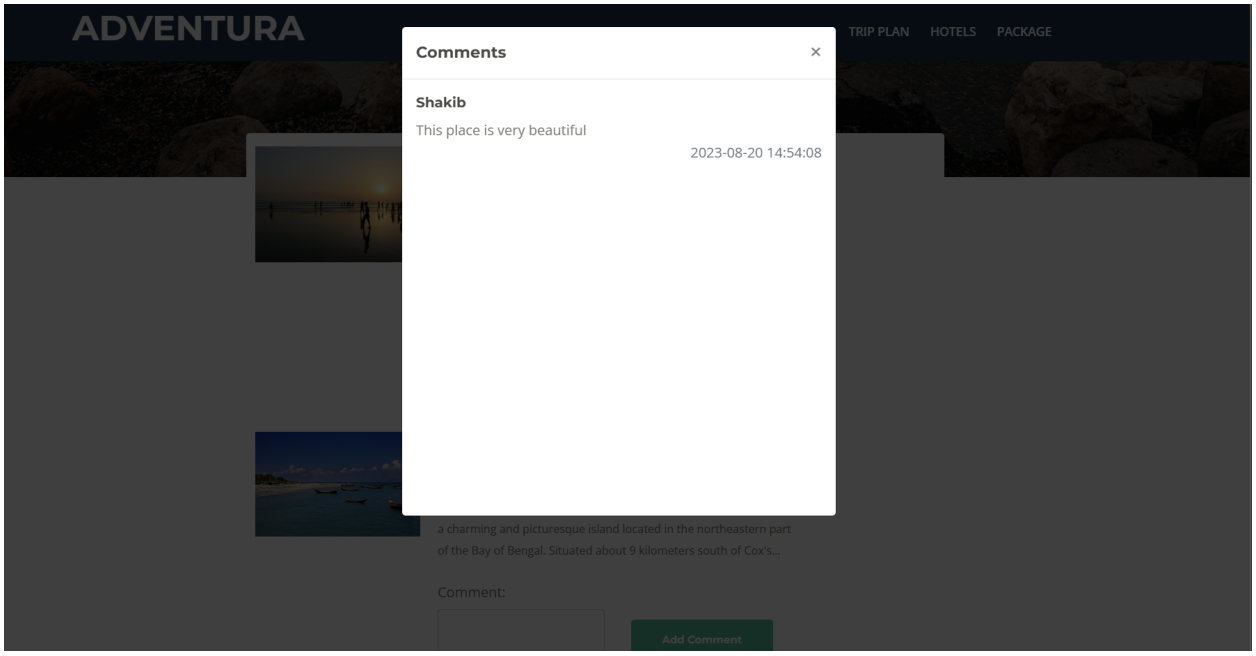
Main purpose of this page-

- **Search option:** Here the user can search for a specific spot directly with the spot name or even with the district name. The search bar will provide the user with the spot name with a glimpse of necessary information.
- **Information box:** Through the search bar or automatically we will get the glimpse of the spots. If we press on the name of a particular spot, we will get a description box which will provide necessary details about the place and show the picture as well.
- **Comment option:** Users can also comment here individually for the spots.






ADVENTURA
PACKAGE
Saint Martin Island
Disctrict
Cox's Bazar
Transports
Greenline(Ship)
Description
Saint Martin Island, also known as Narikel Jinjira (Coconut Island), is a charming and picturesque island located in
Close
Comment:
Add Comment


ADVENTURA
TRIP PLAN
HOTELS
PACKAGE
Comments
Shakib
This place is very beautiful
2023-08-20 14:54:08
a charming and picturesque island located in the northeastern part
of the Bay of Bengal. Situated about 9 kilometers south of Cox's...
Comment:
Add Comment

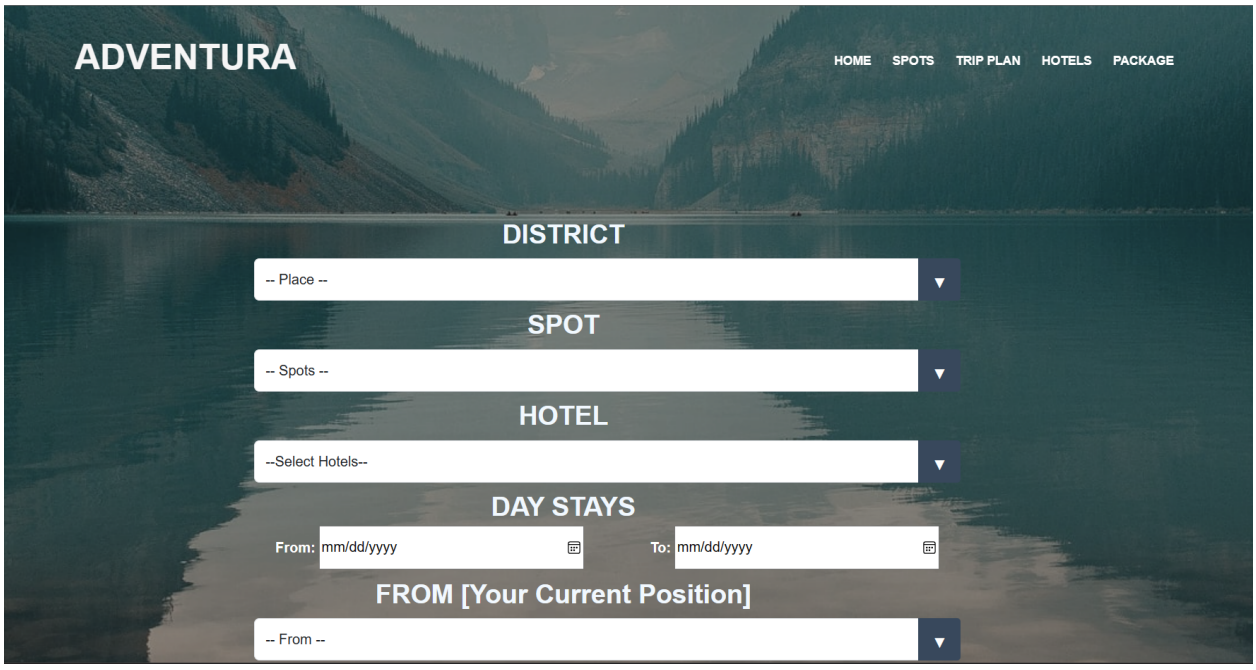
Trip Plan

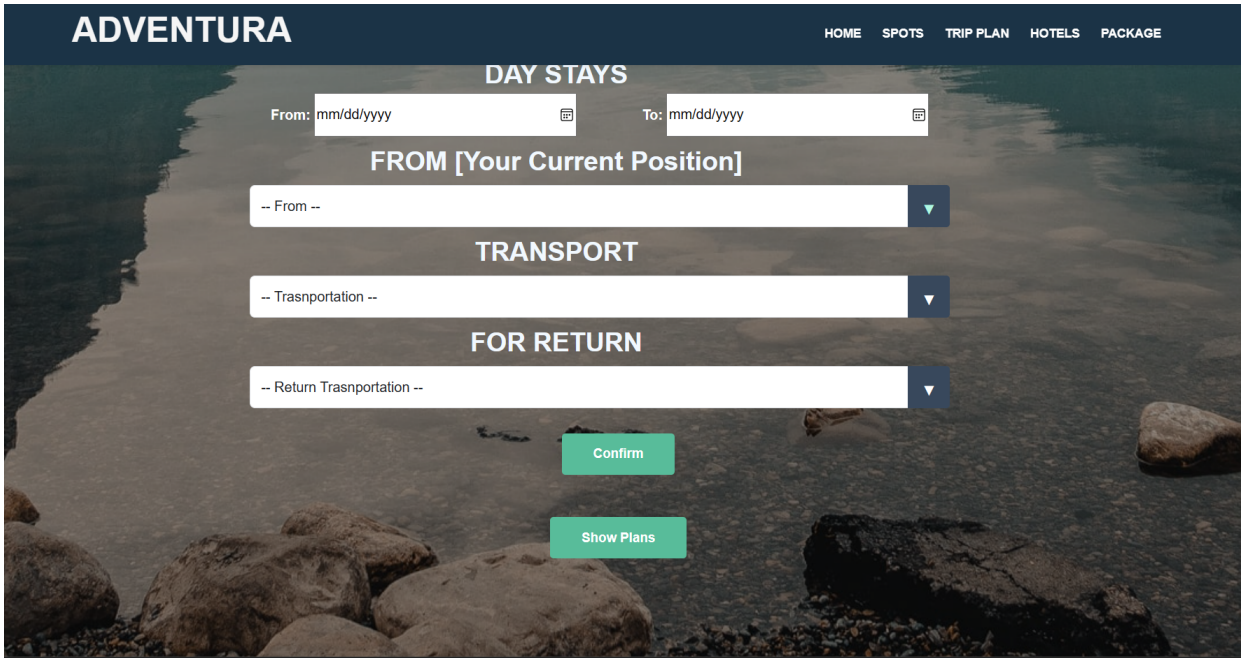
Main purpose of this page-

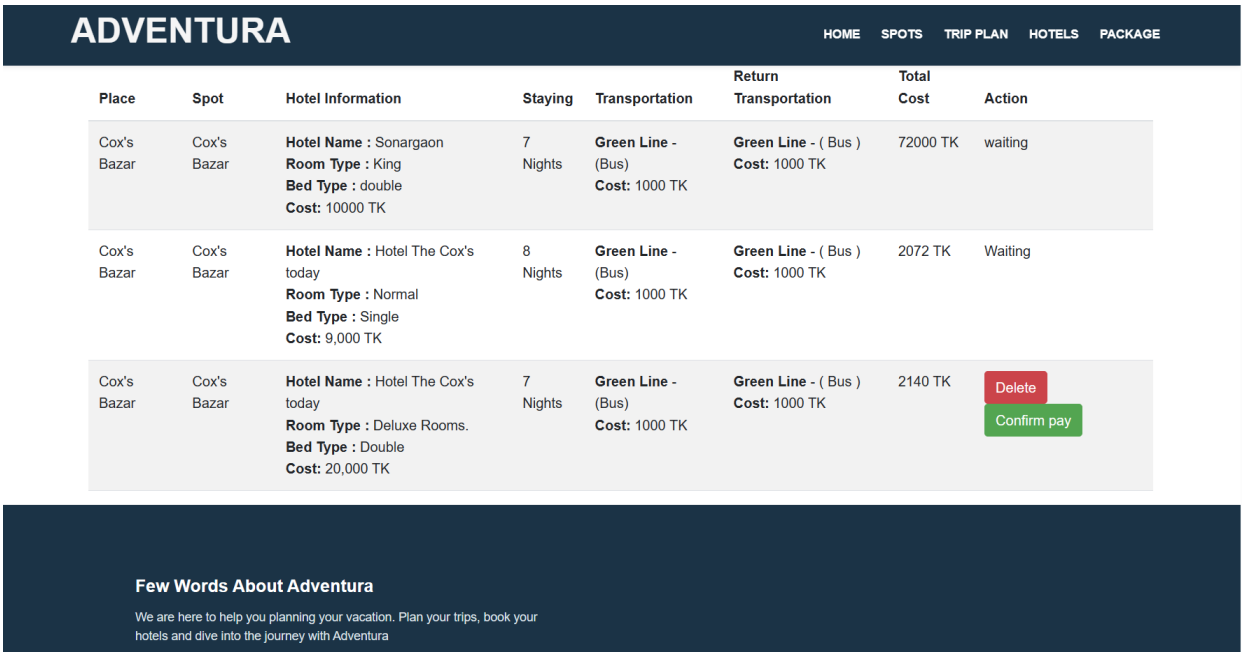
- **Choose district:** From the drop down, users can choose the district according to their preference.
- **Choose spot:** According to the district preference, user will see the particular spot options of that district.
- **Choose hotel:** This cell will show the relevant hotel/resort options for that particular spot.
- **Day Stays:** From the calendar cell the user will be able to select the dates and will count the days accordingly.
- **Current Position:** From this drop down user will choose the place from where he is to start the journey.
- **Transport:** Here, the user will choose the way of traveling towards the destination spot.
- **For Return:** This option is to manage the returning way.

Two buttons of this page-

- **Confirm:** To confirm the plan this button will provide the user with the overall plans. From here, users can confirm the plan by payment or can delete this as well.
- **Show Plans:** By clicking this button, users can view the already made plans.





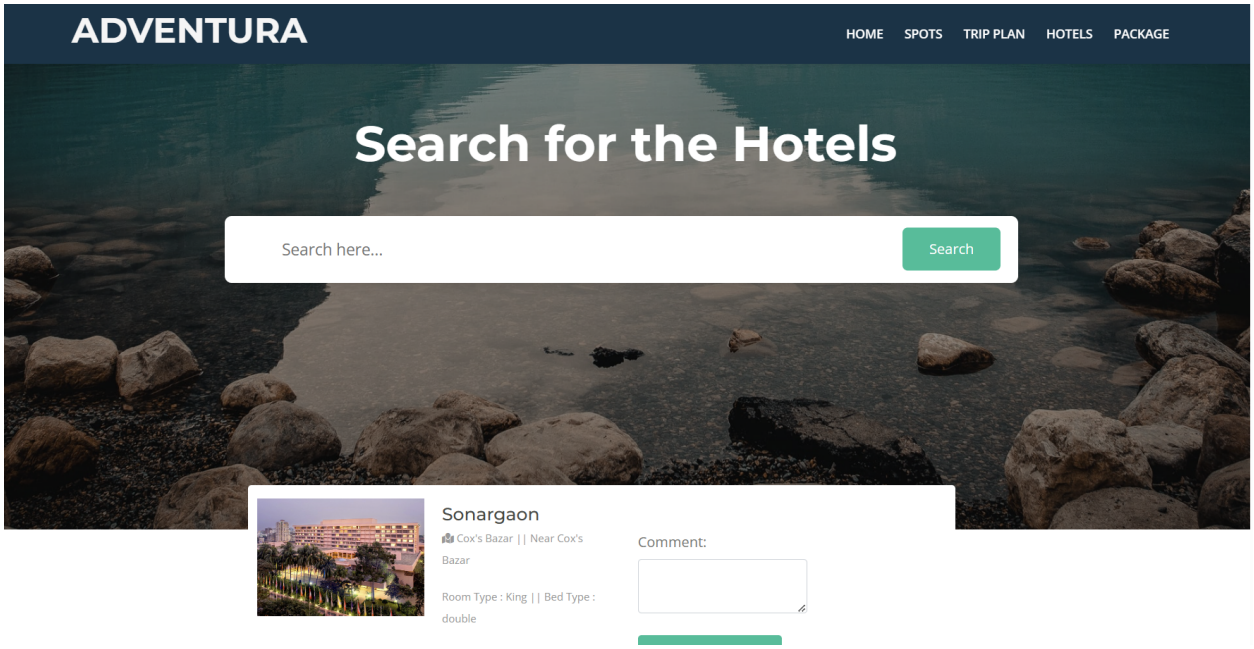


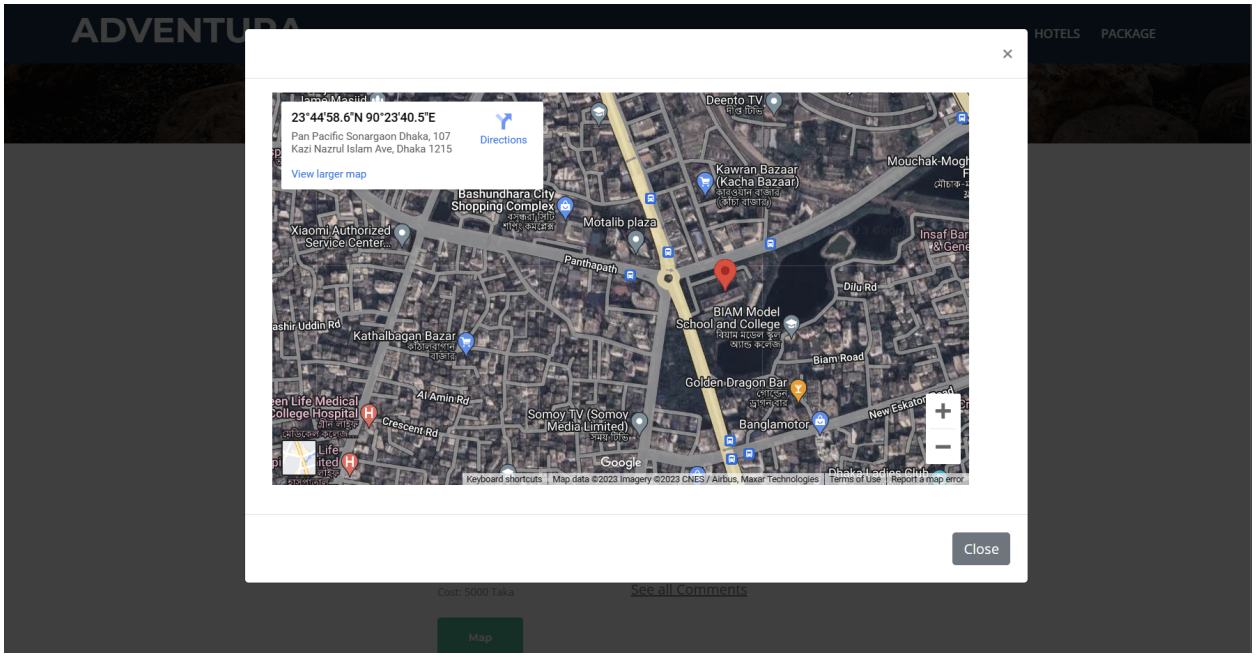
Place	Spot	Hotel Information	Staying	Transportation	Return Transportation	Total Cost	Action
Cox's Bazar	Cox's Bazar	**Hotel Name :** Sonargaon **Room Type :** King **Bed Type :** double **Cost:** 10000 TK	7 Nights	**Green Line** - (Bus) **Cost:** 1000 TK	**Green Line** - (Bus) **Cost:** 1000 TK	72000 TK	waiting
Cox's Bazar	Cox's Bazar	**Hotel Name :** Hotel The Cox's today **Room Type :** Normal **Bed Type :** Single **Cost:** 9,000 TK	8 Nights	**Green Line** - (Bus) **Cost:** 1000 TK	**Green Line** - (Bus) **Cost:** 1000 TK	2072 TK	Waiting
Cox's Bazar	Cox's Bazar	**Hotel Name :** Hotel The Cox's today **Room Type :** Deluxe Rooms. **Bed Type :** Double **Cost:** 20,000 TK	7 Nights	**Green Line** - (Bus) **Cost:** 1000 TK	**Green Line** - (Bus) **Cost:** 1000 TK	2140 TK	Delete Confirm pay

Hotels

Main purpose of this page-

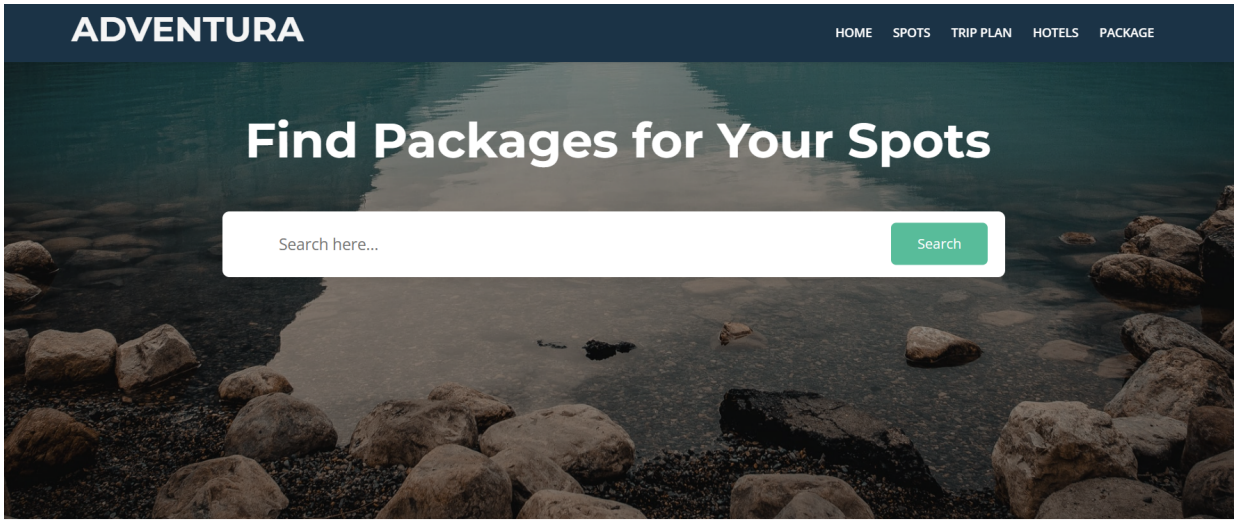
- **Search option:** We can search for particular hotels/resorts here.
- **Comment option:** Users can also comment here individually for the hotels.
- **Map navigation:** Precise location of the resorts will be shown.



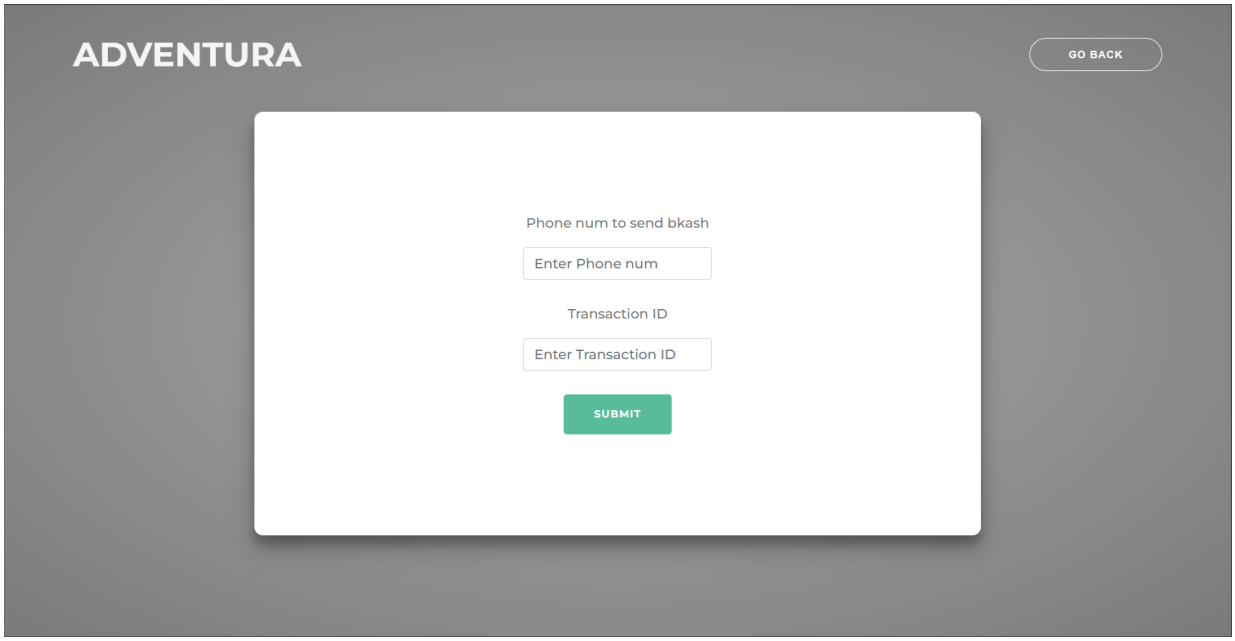


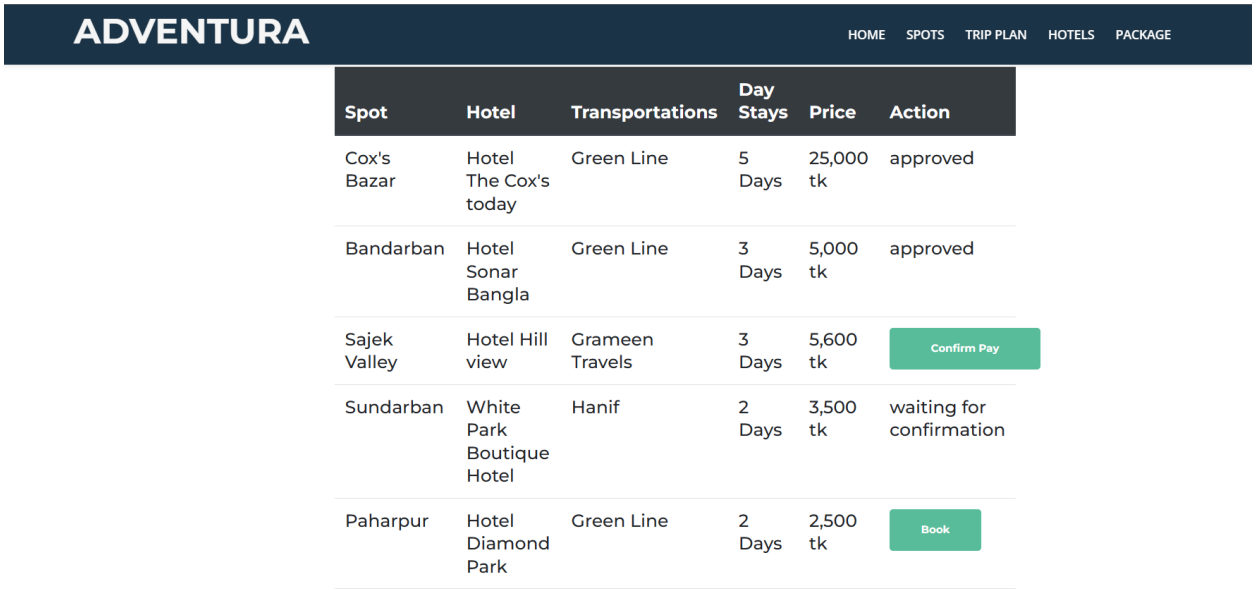
Package

Here users will get ready and plan a package for a particular spot. In the search icon users are to put the spot name and will get the offered packages. Some ready tour plans will be offered by the admin for the user.



Spot	Hotel	Transportations	Day Stays	Price	Action
Cox's	Hotel	Green Line	5	25,000	approved

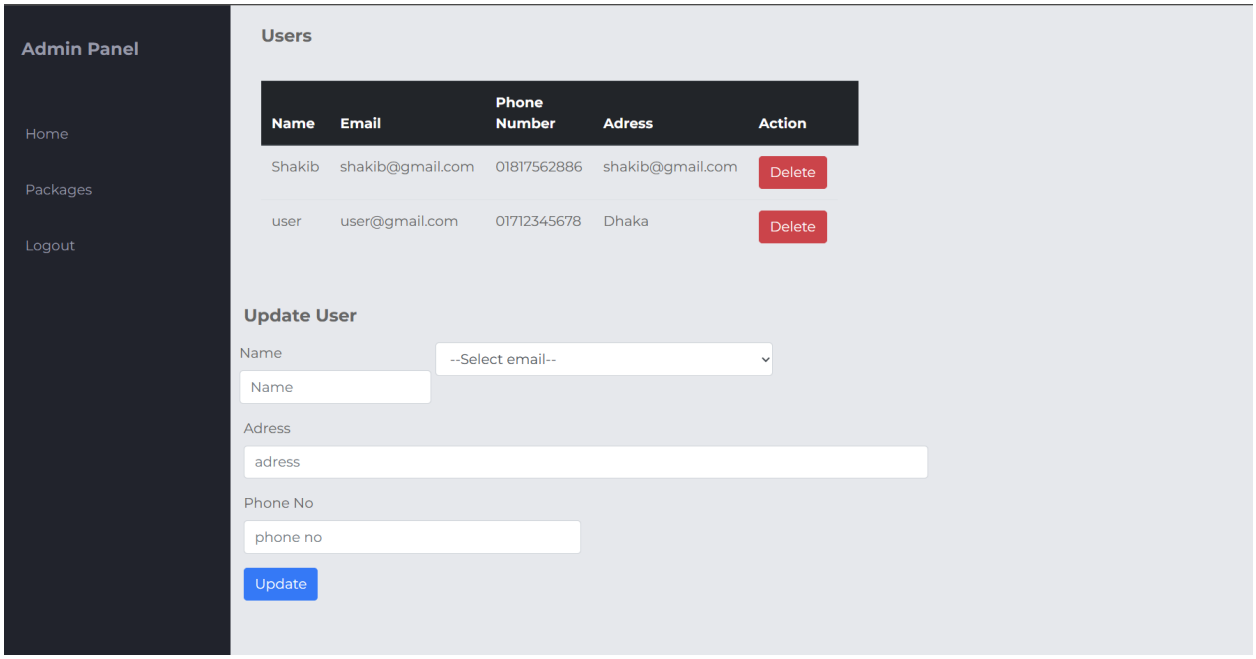


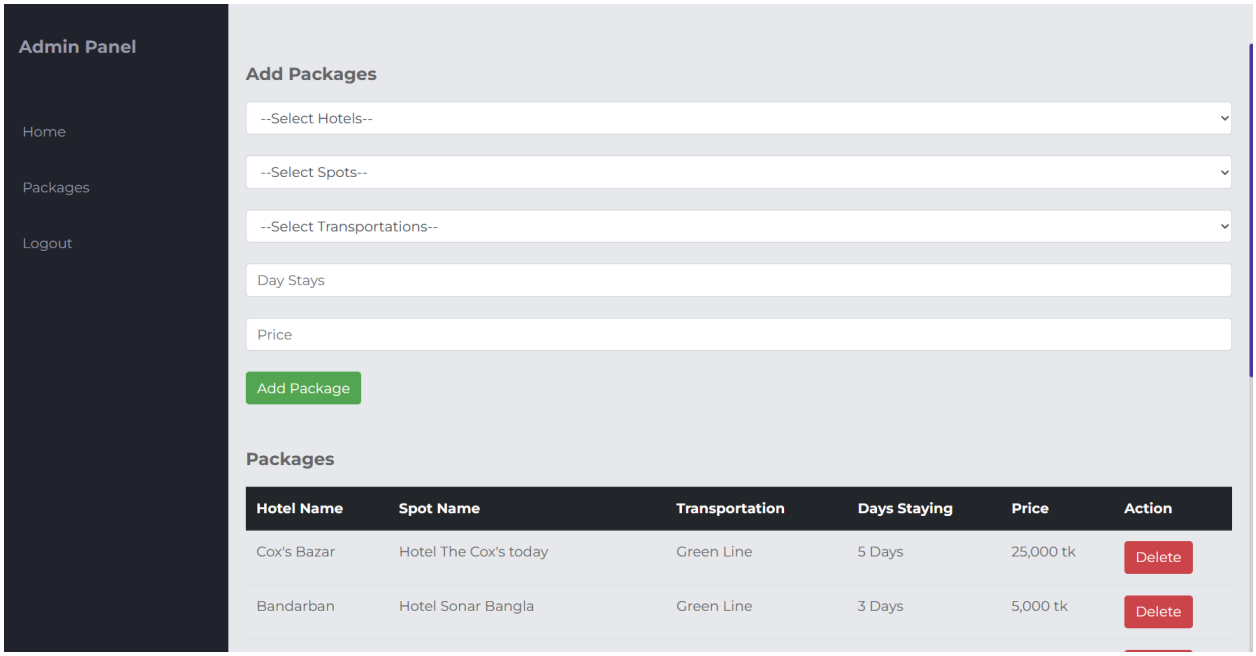

ADVENTURA
HOME
SPOTS
TRIP PLAN
HOTELS
PACKAGE
Spot
Hotel
Transportations
Day Stays
Price
Action
Cox's Bazar
Hotel The Cox's today
Green Line
5 Days
25,000 tk
approved
Bandarban
Hotel Sonar Bangla
Green Line
3 Days
5,000 tk
approved
Sajek Valley
Hotel Hill view
Grameen Travels
3 Days
5,600 tk
Confirm Pay
Sundarban
White Park Boutique Hotel
Hanif
2 Days
3,500 tk
waiting for confirmation
Paharpur
Hotel Diamond Park
Green Line
2 Days
2,500 tk
Book

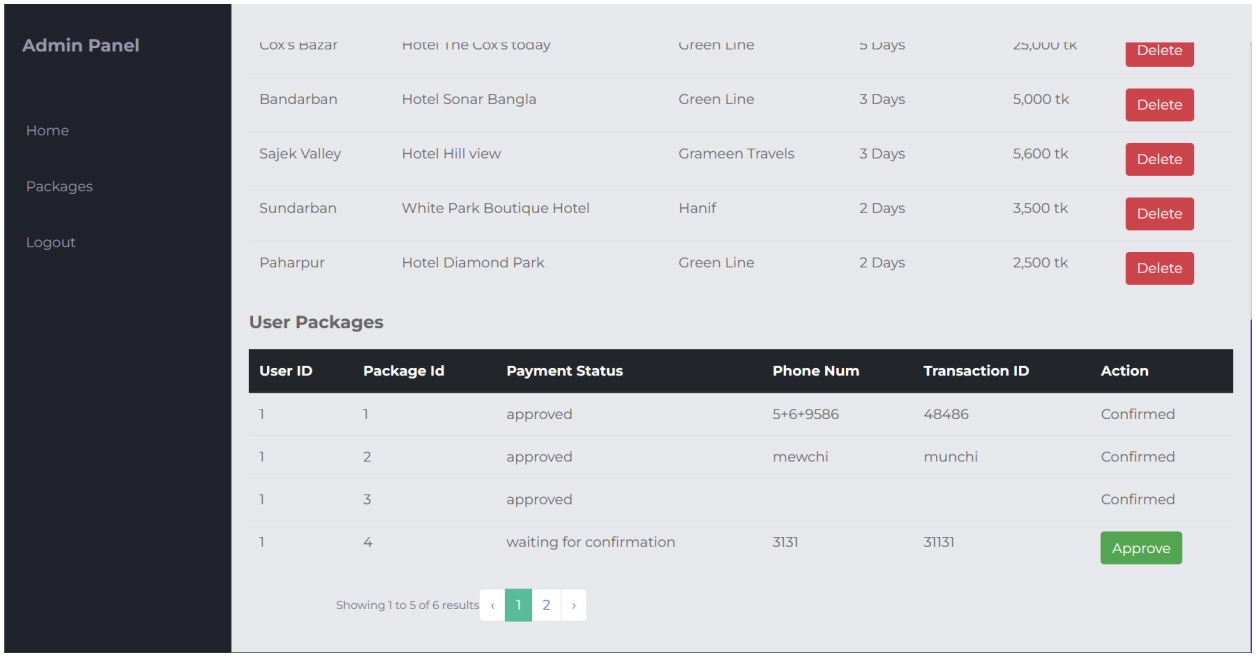
Admin Interface

Main purpose of the Admin user-

- **Users:** Admin can see all the verified users and also can remove the access of them by deleting their information.
- **Update User:** Admin will also be able to update any information of the users and thus save it. By the registered email, the admin can fetch or update.
- **Add Package:** Admin can create packages and offer them to the user.
- **Delete Package:** Deleting any existing package can be done as well by admin
- **Approve User Package:** After completion of the payment by the user, admin can approve it.




Admin Panel
Home
Packages
Logout
Add Packages
--Select Hotels--
--Select Spots--
--Select Transportations--
Day Stays
Price
Add Package
Packages
Hotel Name
Spot Name
Transportation
Days Staying
Price
Action
Cox's Bazar
Hotel The Cox's today
Green Line
5 Days
25,000 tk
Delete
Bandarban
Hotel Sonar Bangla
Green Line
3 Days
5,000 tk
Delete


Admin Panel
Home
Packages
Logout
Cox's Bazar
Hotel The Cox's today
Green Line
5 Days
25,000 tk
Delete
Bandarban
Hotel Sonar Bangla
Green Line
3 Days
5,000 tk
Delete
Sajek Valley
Hotel Hill view
Grameen Travels
3 Days
5,600 tk
Delete
Sundarban
White Park Boutique Hotel
Hanif
2 Days
3,500 tk
Delete
Paharpur
Hotel Diamond Park
Green Line
2 Days
2,500 tk
Delete
User Packages
User ID
Package Id
Payment Status
Phone Num
Transaction ID
Action
1
1
approved
5+6+9586
48486
Confirmed
1
2
approved
mewchi
munchi
Confirmed
1
3
approved
Confirmed
1
4
waiting for confirmation
3131
31131
Approve
Showing 1 to 5 of 6 results
1
2

Frontend Development

Front-end development for this Laravel project involves creating the user interface, user experience, and interactive components that users interact with directly. Laravel, being a backend framework, works in conjunction with frontend technologies to deliver a complete web application. Here's how frontend development is typically approached in a Laravel project:

1. Blade Templating: Views in Laravel are responsible for presenting data to users. The Blade templating engine allows you to embed PHP code within HTML templates, making it easier to create dynamic and reusable views. Here is a blade template for a trip plan where the total cost is calculated by the parameter of hotel booking or not hotel booking.

```
@if(count($all_plan)!=0)

    @foreach($all_plan as $plan)
    @php
        $return_hotel_parts = explode(' - ', $plan->hotel_info);
        $total_parts = count($return_hotel_parts);

        if (is_null($return_hotel_parts)){
            $total_cost =  floatval($plan->tcost) +  floatval($plan->rtcost);
        }else{
            $total_cost =  ($plan->dayStays*(floatval($return_hotel_parts[$total_parts-1]))) +  floatval($plan->tcost) +  floatval($plan->rtcost);
        }

    @endphp
    <tr>

        <td>{{$plan->dis_name}}</td>
        <td>{{$plan->spot_name}}</td>
        @if(!empty($plan->hotel_info))
            <td>

                @for ($i = 0; $i < $total_parts; $i++)
                    @if ($i == 0)
                        <b>Hotel Name :</b>  {{ $return_hotel_parts[$i] }}<br>
                    @elseif ($i == 1 )
                        <b>Room Type :</b>   {{ $return_hotel_parts[$i] }}<br>
                    @elseif ($i == 2 )
                        <b>Bed Type :</b>  {{ $return_hotel_parts[$i] }}<br>
                    @else
                        <b>Cost:</b>   {{ $return_hotel_parts[$i] }} TK<br>
                    @endif
                @endfor
            </td>
            <td>{{$plan->dayStays}} Nights</td>
        @else
            <td></td>
            <td></td>
        @endif
```

Here is another ajax implementation for creating connection with the controller in spot.blade.php file.

```
$(document).ready(function() {

    $('.detail-btn').click(function() {
        const id = $(this).attr('data-id');

        $.ajax({
            url: 'spot_pop/'+id,
            type: 'GET',
            data: {
                "id": id
            },
        success:function(data) {


            $('#spot-title').html(data[0].name);
            $('#spot-img').attr('src', 'store_pics/'+data[0].pictures);
            $('#spot-dist').html(data[0].disName);
            $('#spot-desc').html(data[0].description);

           const groupedTransports = {};
            for (const item of data) {
                const key = item.transportName;
                if (groupedTransports[key]) {
                    groupedTransports[key].push(item.transport_type);
                } else {
                    groupedTransports[key] = [item.transport_type];
                }
            }
```

2. HTML, CSS, and JavaScript: HTML (Hypertext Markup Language), CSS (Cascading Style Sheets), and JavaScript are fundamental building blocks of frontend development. They determine the structure, layout, styling, and interactivity of web pages. Here is a screenshot of authentication page CSS:

```
@import url('https://fonts.googleapis.com/css?family=Montserrat:500,700&display=swap&subset=latin-ext');

* {
  box-sizing: border-box;
}

body {
  background: #f6f5f7;
  display: flex;
  justify-content: center;
  align-items: center;
  flex-direction: column;
  font-family: 'Montserrat', sans-serif;
  height: 90vh;
  margin: -20px 0 50px;
  margin-top: 40px;
}

h1 {
  font-weight: bold;
  margin: 0;
}

h2 {
  text-align: center;
}

p {
  font-size: 14px;
  font-weight: 100;
  line-height: 20px;
  letter-spacing: 0.5px;
  margin: 20px 0 30px;
}
```

And here is a CSS of admin panel sidebar:

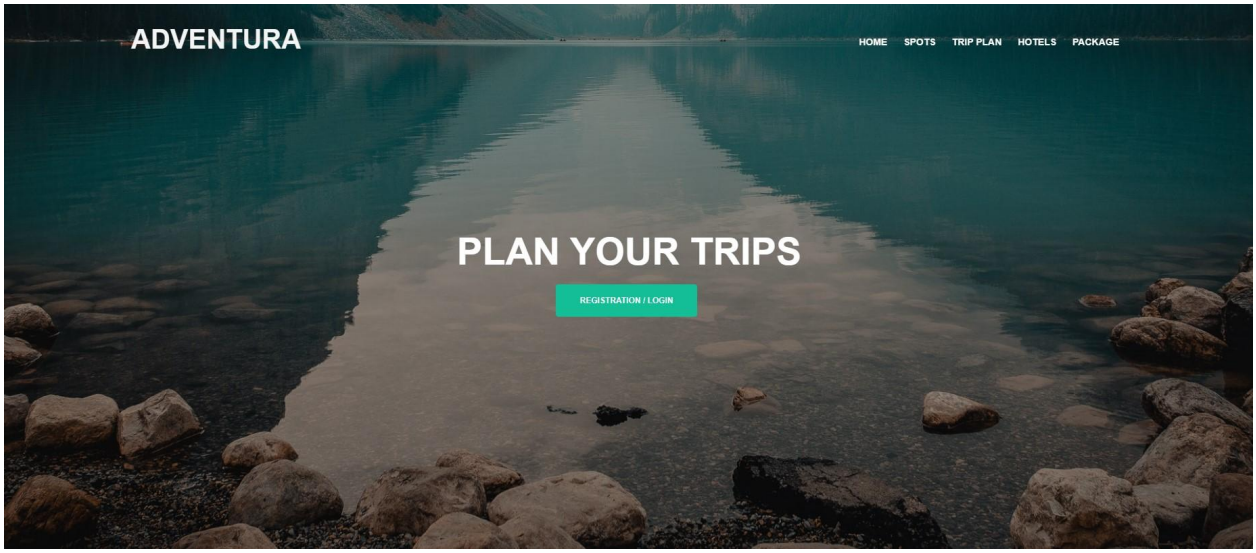
```css
#sidebar {
  grid-area: sidebar;
  height: 100%;
  background-color: #21232d;
  color: #9799ab;
  overflow-y: auto;
  transition: all 0.5s;
  -webkit-transition: all 0.5s;
}

.sidebar-title {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 20px 20px 20px 20px;
  margin-bottom: 30px;
}

.sidebar-title > span {
  display: none;
}

.sidebar-brand {
  margin-top: 15px;
  font-size: 20px;
  font-weight: 700;
}
```

3. UX and UI Design: Focus on creating intuitive user experiences (UX) and visually appealing user interfaces (UI). This involves wireframing, prototyping, and designing user flows to optimize user interactions. Here is the homepage UI of our project:



Backend Development

We use laravel framework as our backend development. Laravel is a powerful and popular PHP web framework that provides tools, libraries, and conventions for building robust, maintainable, and scalable web applications. The backend of a web application, including its data management, business logic, and server-side operations, can be developed using Laravel. Those functionality comes mostly from Laravel built in features. Here some of the major backend features of Laravel:

1. Routing: Laravel offers a robust routing system that allows you to define routes and map them to specific controllers and actions. This is essential for handling different HTTP requests and directing them to the appropriate parts of your backend code. Here in this picture, we showed the web.php route.

routes > web.php

```
22
23  Route::get('/',[HomePage::class,'imageAndDescription'])->middleware('islogout');
24
25  Route::get('/login',[AuthController::class,'login'])->middleware('islogout');
26  Route::post('login-user',[AuthController::class,'loginUser'])->name('login-user');
27
28  Route::get("/register",[AuthController::class,'register'])->middleware('islogout');
29  Route::post("/register-user",[AuthController::class,'registerUser']) -> name('register-user');
30
31  Route::get("/dashboard",[AuthController::class,'dashboard'])->middleware('isLoggedIn');
32
33  Route::get('/logout',[AuthController::class,'logout']);
34
35  Route::get('/spots',[Spots::class,'spot_page'])->middleware('isLoggedIn');
36  Route::get('spot_pop/{id}',[Spots::class,'spot_pop'])->name('spot_pop');
37  Route::post('add_comment/',[Spots::class,'add_comment'])->name('add_comment');
38  Route::get('view_comment/{id}',[Spots::class,'view_comment'])->name('view_comment');
39
40
41
42  Route::get('/hotels',[hotels::class,'hotel_page'])->middleware('isLoggedIn');
43  Route::get('map_pop/{id}',[hotels::class,'map_pop'])->name('map_pop');
44  Route::get('view_comment_hotel/{id}',[hotels::class,'view_comment_hotel'])->name('view_comment_hotel');
45  Route::post('add_comment_hotel/',[hotels::class,'add_comment_hotel'])->name('add_comment_hotel');
46
```

2. Controllers: Controllers in Laravel contain the application's logic. They receive incoming requests, process data, interact with models, and return appropriate responses. Controllers help separate the business logic from the presentation layer. Here in this screenshot of AuthController.php where we showed some different controllers for multiple functions for login and registration:

```php
class AuthController extends Controller
{
    public function login(){
        return view("auth.login");
    }

    public function loginUser(Request $request){
        $request->validate([

            'email'=>'required|email',
            'pass'=>'required|min:6|max:12'

        ]);

        $user = DB::table('users')->where('email', $request->email)->first();



        if ($user){
            $type = DB::table('users')->where('email', $request->email)->where('user_type', 'user')->select('user_type')->first();

            if ($type!=null){
                if ($type->user_type == 'user' ){
                    if (Hash::check($request->pass, $user->pass)){

                        $request->session()->put('loginId', $user->id);
                        return redirect('dashboard');

                    }else{
                        return back()->with('fail','Password not match or do not have any email');
                    }
                }
            }elseif($type == null){
                if ($request->pass == $user->pass){

                    $request->session()->put('loginId', $user->id);
                    return redirect('adminDashboard');

                }else{
                    return back()->with('fail','Password not match or do not have any email');
                }
```

Also, this is a screenshot of admin controller for package updates:

```php
    public function package(){
        $package =DB::table('packages')
            ->join('spots','spots.id','=','packages.spots_id')
            ->join('hotels','hotels.id','=','packages.hotels_id')
            ->join('transportations','transportations.id','=','packages.transportations_id')
            ->leftjoin('user_packages','user_packages.packages_id','=','packages.id')
            ->select('packages.*','spots.name as spotName','spots.id as spotId','hotels.name as hotelName','transportations.transport_name as transport_name'
            ->paginate(5);
        $hotel = DB::table('hotels')
        ->get();
        $spot= DB::table('spots')
        ->get();
        $transport= DB::table('transportations')
        ->join('spots','spots.id','=','transportations.spots_id')
        ->join('districts','districts.id','=','transportations.districts_id')
        ->select('transportations.*','districts.name as disName','spots.name as spotName' )
        ->get();
        $packagess  = DB::table('user_packages')->leftjoin('packages','packages.id','=','user_packages.id')->select('user_packages.*')->get();

        return view("adminDashboard.package",compact('package','hotel','spot','transport','packagess'));
    }//

    public function destroy_package($id){

        $data = DB::table('packages')
        ->where('id','=',$id )
        ->delete();

        if($data){
            return redirect('adminPackage')->with('deleted','Data has been deleted');
        }
    }
```

3. Database Migrations: Laravel's migration system allows you to version-control your database schema changes using code. Migrations make it easy to create, modify, and rollback database tables and relationships. Here is the migration of the Spots database of our project.

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('spots', function (Blueprint $table) {
            $table->id();
            $table->string('name');
            $table->string('description');
            $table->string('pictures');
            $table->string('longitude and latitude');
            $table->foreignId('districts_id')->constrained()->cascadeOnDelete();

        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('spots');
    }
};

```

4. Middleware: Middleware in Laravel provides a way to filter HTTP requests entering your application. It's commonly used for tasks like authentication, authorization, and request preprocessing. Here is a middleware functionality for login sessions.

```php
<?php

namespace App\Http\Middleware;

use Closure;
use Illuminate\Http\Request;
use Symfony\Component\HttpFoundation\Response;

class AuthCheck
{
    /**
     * Handle an incoming request.
     *
     * @param  \Closure(\Illuminate\Http\Request): (\Symfony\Component\HttpFoundation\Response)  $next
     */
    public function handle(Request $request, Closure $next): Response
    {
        if (!Session()->has('loginId')){

            return redirect('login')->with('fail','You have to login first');

        }
        return $next($request);
    }
}

```

Technology

Framework: Laravel

We implemented laravel for our project as this is the best framework for PHP and also very easy to use and implement. Laravel is a popular open-source PHP web framework used for building web applications and websites. It was created by Taylor Otwell and first released in 2011. Laravel follows the Model-View-Controller (MVC) architectural pattern, which separates the application logic into different components to enhance code organization, reusability, and maintainability.

Here's a breakdown of the key components and features of Laravel:

1. Routing: Laravel provides a powerful routing system that allows developers to define how application URLs should be handled, mapping them to specific controllers and actions.
2. Model-View-Controller (MVC): This architectural pattern separates the application into three components: Models (representing the data and database interactions), Views (handling the presentation layer), and Controllers (managing the application's logic).
3. Artisan CLI: Laravel includes a command-line tool called Artisan, which offers various commands to automate common development tasks, such as generating code, migrating databases, and more.

4. Database Migration: Laravel's migration system enables developers to define database schema changes using code, making it easy to version-control and share database structure changes among team members.

5. Blade Templating Engine: Blade is a lightweight templating engine that allows developers to write dynamic, reusable templates with PHP code embedded directly within the HTML markup.

6. Middleware: Laravel's middleware feature allows you to filter HTTP requests entering your application. This is useful for tasks such as authentication, authorization, and request preprocessing.

7. Authentication and Authorization: Laravel provides built-in authentication and authorization functionalities, making it simpler to handle user authentication and manage access control to different parts of your application.

8. Routing: Laravel encourages the use of RESTful routing, making it easy to design APIs that adhere to REST principles.

9. Packages and Libraries: Laravel has a vibrant ecosystem of packages and libraries, known as Composer packages, that can be easily integrated into your projects to extend functionality without reinventing the wheel.

Laravel's combination of modern features, elegant syntax, and comprehensive documentation has contributed to its popularity among web developers. It's widely used for building a variety of web applications, from small projects to large-scale, enterprise-level systems.

Language: PHP

PHP and Laravel are closely related technologies that work together to enable the development of dynamic and feature-rich web applications.

1. PHP as the Foundation: Laravel is built upon the foundation of PHP, which serves as the scripting language that powers the dynamic aspects of web applications. PHP allows developers to embed server-side code within HTML, generating content based on user interactions, database queries, and other backend processes.

2. Server-Side Execution: Laravel, as a web framework, uses PHP to execute server-side logic. When a user requests a web page or performs an action on a Laravel-powered application, PHP processes the request, interacts with databases, performs calculations, and generates the dynamic content to be sent to the user's browser.

3. Embedding PHP in Blade Templates: Laravel's Blade templating engine allows developers to embed PHP code within HTML templates, facilitating dynamic content rendering. This integration of PHP within Blade templates enhances the separation of concerns and improves code readability.

4. Database Integration: Laravel leverages PHP to connect to various databases, execute SQL queries, and manage data interactions. PHP's ability to interact with databases like MySQL and PostgreSQL is essential for creating database-driven applications using Laravel.

5. Laravel Framework: Laravel is a PHP-based web framework that simplifies the development process by providing a set of tools, libraries, and conventions. It offers features like routing, database migrations, authentication, and templating, all of which are implemented using PHP code.

6. Composer and PHP Dependencies: Composer, a PHP package manager, is commonly used in the Laravel ecosystem. It allows developers to manage external libraries and dependencies, making it easier to integrate third-party components into Laravel projects.

PHP's ability to seamlessly generate dynamic content and its widespread adoption across the web have solidified its role as one of the most important technologies for web development. Many popular websites and applications, including content management systems like WordPress and e-commerce platforms like Magento, rely on PHP to deliver interactive and engaging user experiences.

Github Repository

Link: https://github.com/J0vial/Adventura-Plan-Your-Trips